

1. Project Overview

This project analyzes global BMW vehicle sales trends and the factors that influence them using the BMW Global Automotive Sales Dataset (2018–2025). The dataset contains monthly records of vehicle sales across different regions, models, and economic conditions, including variables such as units sold, average vehicle price, revenue, battery electric vehicle (BEV) share, GDP growth, and fuel price index. Within the automotive market analytics domain, the goal of this project is to support exploratory data analysis through an interactive visualization dashboard that allows users to examine regional sales patterns, compare vehicle model performance, explore the growth of electric vehicle adoption, and investigate how economic indicators relate to BMW sales performance.

2. Research Questions

1. How have BMW vehicle sales evolved across global regions over time?

Mid-level questions

- 1a. How do BMW vehicle sales change over time within each region?

Analyze -> units sold over time within regions

- 2a. How do BMW vehicle sales compare across regions?

Analyze -> units sold across regions

Low-level questions

- 1b. Which region has the highest BMW vehicle sales?

Abstract Task

1c. *Find Extremum* -> *units sold across regions*

- 2b. What is the total number of vehicles sold in each region?

Abstract Task

2c. *Compute Derived Value* -> *total units sold per region*

2. How have BMW vehicle sales evolved across global regions over time?

Mid-level questions

1a. What relationship exists between BEV share and total BMW sales?

Analyze -> BEV share and total units sold

2a. How has BEV share evolved over time?

Analyze -> BEV share over time

Low-level questions

1b. What is the average BEV share for each year?

Abstract Task

1c. *Compute Derived Value -> average BEV share per year*

2b. What is the range of BEV share values across the dataset?

Abstract Task

2c. *Determine Range -> BEV share values*

3. How does the adoption of battery electric vehicles (BEV share) relate to BMW sales performance?

Mid-level questions

1a. How does revenue vary across BMW vehicle models?

Analyze -> revenue across vehicle models

2a. Which BMW models account for the highest number of units sold?

Search -> units sold across vehicle models

Low-level questions

1b. What is the total number of units sold for each BMW model?

Abstract Task

1c. *Compute Derived Value -> total units sold per vehicle model*

2b. Which BMW model has the highest sales volume?

Abstract Task

2c. Find Extremum -> units sold across vehicle models

4. How do different BMW vehicle models contribute to total sales and revenue?

Mid-level questions

1a. How does fuel price variation relate to BMW vehicle sales?

Analyze -> relationship between fuel price indexes and BMW vehicle sales

2a. What relationship exists between GDP growth and BMW vehicle sales?

Analyze -> relationship between GDP growth and BMW vehicle sales

Low-level questions

1b. What is the GDP growth value for each time period?

Abstract Task

1c. Retrieve Value -> GDP growth for each time period

2b. How do BMW sales differ between periods of higher and lower GDP growth?

Abstract Task

2c. Compare -> BMW vehicle sales across GDP growth levels